

TABLE OF CONTENTS

I-002 LOCAL CONTROL PANELS

	Page No.
1. SCOPE OF WORK.....	I 002-3
2. CODES AND STANDARD	I 002-3
3. ENVIRONMENTAL REQUIREMENTS	I 002-4
4. DETAILED REQUIREMENTS	I 002-4
4.1 General.....	I 002-4
4.2 Panel Function and Special Requirements.....	I 002-6
4.2.1 Local Control Panels (Process)	I 002-6
4.2.2 Local Control Panels (Electrical)	I 002-6
4.2.3 System Cabinets	I 002-7
4.2.4 I/O cabinets	I 002-7
4.2.5 Termination Cabinet.....	I 002-7
4.2.6 Marshalling Cabinet	I 002-8
4.3 Construction	I 002-8
4.4 Painting and Finishing	I 002-9
4.4.1 General.....	I 002-9
4.4.2 Finishing for Indoor Panels.....	I 002-10
4.4.3 Finishing for Outdoor Panels	I 002-10
4.5 Door, Opening and Covers.....	I 002-10
4.6 Terminal Blocks	I 002-11
4.6.1 Terminals for Power Wiring	I 002-12
4.6.2 Terminals for Instrument Wiring	I 002-12
4.7 Wiring	I 002-12

4.7.1	Wire	I 002-12
4.7.2	Wiring Arrangement.....	I 002-13
4.7.3	Wiring Termination.....	I 002-14
4.8	Name Plates.....	I 002-15
4.9	Panel Mounted Devices	I 002-15
4.9.1	General.....	I 002-15
4.9.2	Annunciators	I 002-16
4.9.3	Relay Protection and Monitoring Equipment	I 002-18
4.9.4	Indicating Lamps.....	I 002-18
4.9.5	Control Switches	I 002-18
4.9.6	Push Buttons, and Push Button Station	I 002-18
4.9.7	Sound Power Jack System	I 002-19
4.9.8	Auxiliary instrument Transformer	I 002-19
4.10	Interior Lighting	I 002-19
4.11	Convenience Plug.....	I 002-19
4.12	Grounding	I 002-19
4.13	Ventilation Fans	I 002-20
4.14	Anti Condensation Heaters	I 002-20
4.15	Piping and Pneumatic Connection.....	I 002-20
4.16	Hazardous Location.....	I 002-21
4.17	Surge with Stand Capability.....	I 002-21
5.	TESTING	I 002-21
6.	QUALITY ASSURANCE /QUALITY CONTROL	I 002-23

I 002 LOCAL CONTROL PANELS

1. SCOPE OF WORK

This Technical Specification defines the general requirements for Local Control Panels.

2. CODES AND STANDARD

The Contractor shall comply with the requirements of Part 4, Subsection 2.5” Codes and Standard”. The following Codes and Standard of the following organizations are especially applicable to the design, manufacturing, and testing of the equipment included in this Technical Specification.

ANSI B31.1	Code of Pressure Piping
ANSI C37.90a	Guide for Surge Withstand Capability Test
ASTM A366	Steel, Carbon, Cold Rolled Sheet
IEC 60050	International Electro-technical Vocabulary
IEC 60079	Electrical apparatus for explosive gas atmospheres
IEC 60081	Tubular fluorescent lamps for general lighting service
IEC 60082	Ballast for tubular fluorescent lamps
IEC 60083	Plugs and socket-outlet for domestic and similar general use
IEC 60113	Diagrams, charts, and tables
IEC 60117	Recommended Graphical Symbols
IEC 60228	Conductors of insulated cables
IEC 60245	Rubber insulated cable of rated voltage up to and including 450/750 Volts
IEC 60309	Plugs, socket-outlets and couplers industrial purposes
IEC 60331	Fire-resisting characteristic of electrical cables
IEC 60445	Identification of apparatus terminals and general rules for a uni form system of terminal marking, using and alphanumeric notation
IEC 60446	Identification of insulated and bare conductors by colors
IEC 60478	Stabilized power supplies, DC output
IEC 60529	Classification of degree of protection provided by enclosures

IEC 60598	Luminaries
NFPA70	National Fire Protection Association (National Article 500 Electrical Code), Hazardous (classified) location

3. ENVIRONMENTAL REQUIREMENTS

The local control panels shall be suitable for operation in a Combined Cycle power plant near the sea coast in the tropical climate.

4. DETAILED REQUIREMENTS

4.1 General

This Technical Specification shall be applicable to all types of local control panels, cabinets, electrical enclosures, process control panel, etc. In general these panels shall be used to group, support and enclose control and indicating devices required for the control of specific systems or equipment.

All local control panels, cabinets and enclosures shall be furnished complete with all devices, mounted, wired and piped in the factory before being shipped to the Site. Unless otherwise specified or as agreed by the Owner, process fluids shall be excluded from local control panels and confined to areas external to the panels. Only pneumatic and electrical signal lines and air supply lines shall enter the panels.

Contractor shall install and wire equipment in a workmanlike and satisfactory manner as herein specified.

The design of Instrumentation and Control systems and devices shall comply with the requirements as detailed in Part 4, Technical Specification I-001 “Instrumentation and Control” and as detailed herein specified.

The design of Electrical equipment, power supplies, protection relays, control switches, indicating lights and other electrical devices shall comply with the requirements as detailed in Part 4, Technical Specification E-001 “General Electrical Specification” and as detailed herein.

Local control panels shall have the following degree of protection as defined by IEC 60529:

- a. IP 31 - for indoor in an air conditioning room
- b. IP 51 - for indoor in an enclosed room
- c. IP 53 - for indoor away from fluid piping and tanks
- d. IP 54 - for other indoor location
- e. IPW 55 - for outdoor
- f. IPW 65 - for junction boxes, local mounted instrument including pressure switches, level switches, temperature switches, vibration pick up etc.

All hardware, trim, fasteners, and other accessories shall be made from rust proof material.

Reliable uninterruptible power supply (UPS) Systems 220 V, 50 Hz (nominal) power can be provided for all local control panels when required. Where power supply interruptions cannot be tolerated by the process or equipment the panel serves, a second supply of 220 V (nominal) DC shall be provided. Refer to Part 4, Technical Specification E-001.

All Control circuit and transformer fuses shall be in accordance with. Part 4, Technical Specification E-001.

The Contractor shall furnish all transforming, converting and distribution equipment necessary to provide a stable reliable 110 Volt, 50 Hz power supply and low voltage DC supplies for control equipment within the panel and for powering external devices connected to the panel. The use of 220 V AC for control voltage is unacceptable.

If a remote/local selector switch is provided mounted on a local control panel associated with certain equipment for the emergency, maintenance and testing purposes and the equipment also has remote control capability from the operator station in the CCR, then the remote/local selector switch at the local control panel shall be provided with keylock.

4.2 Panel Function and Special Requirements

These panels shall serve as the control point for various plant processes and pieces of equipment or electrical system e.g. water treatment plant, condenser cleaning system, etc.

Process or electrical local control panels shall be free standing, floor mounted, completely enclosed and dust tight, with instruments, relays, gauges, control switches etc. mounted on the front or within.

This type of panel may also contain sampling and process piping to instrumentation if required. Where a free-standing panel is not justified because of the requirements associated with a certain process or item of equipment and with prior approval by the owner, wall mounted panels and panels mounted on equipment skids will be acceptable.

4.2.1 Local Control Panels (Process)

These panels shall serve as the control point for various plant processes and pieces of equipment.

Process local control panels shall be free standing, floor mounted, completely enclosed and dust tight, with instruments, relays, gauges, control switches, etc. mounted on the front or within. This type of panel may also contain sampling and process piping to instrumentation if required. Where a free-standing panel is not justified because of the requirements associated with a certain process or item of equipment and with prior approval by the Owner, wall mounted panels and panels mounted on equipment skids will be acceptable.

4.2.2 Local Control Panels (Electrical)

These panels shall serve as the control point for various plant electrical systems. Electrical local control panels shall be free-standing, floor mounted, completely enclosed and dust-tight with instruments, control switches, relays, terminal blocks etc. mounted on the front or within the panel. This type of panel shall be free from process and pneumatic piping, i.e. process fluids shall not be piped to instruments mounted on the control panel. Where a free-standing panel is not justified because of the requirements associated with a certain process or item of equipment and with prior approval by the Owner, wall mounted panels or panels mounted on equipment skids will be acceptable.

4.2.3 System Cabinets

These cabinets shall support and enclose solid-state control devices, instrument power supplies, etc. which do not have an operator interface function. System cabinets are generally located in an air conditioned electronic equipment room. These cabinets shall be connected to the control panels and to termination cabinets by prefabricated plug-in cables.

Each system cabinet shall consist of two (2) separate sections, one for equipment in front and one other for sockets for incoming and outgoing cables and terminal blocks for internal wiring in the back. The front and back sections of the cabinet shall be separated with a sub-panel. Each section of the cabinet shall be provided with a separate access door.

4.2.4 I/O Cabinets

These cabinets are used for remote I/O hardware and programmable controllers (if required), which collect process inputs and distribute process outputs.

Each I/O cabinet shall consist of two (2) separate sections, one for I/O modules in front and the other for sockets for incoming and outgoing cables and terminal blocks for internal wiring in the back. The front and back sections of the cabinet shall be separated with a sub-panel. Each section of the cabinet shall be provided with a separate access door.

4.2.5 Termination Cabinet

These cabinets provide a termination point for interface between control cables originating in the field and plug-in cable from I/O cabinets, system cabinets, and control panels. Termination cabinets shall not contain control devices except for auxiliary relays and switching devices. All other control devices shall be contained in control panels or system cabinets. Interconnection between termination cabinets, control panels, system cabinets, and I/O cabinets shall be by prefabricated plug-in cables.

Termination cabinets shall be sized to accommodate all incoming and outgoing cables including plug-in cables.

Front and back access door shall be provided for all termination cabinets.

4.2.6 Marshalling Cabinet

These cabinets provide a collection point and interface for interconnecting cables, which originate from equipment.

Marshalling cabinets shall be sized according to the requirements.

Marshalling cabinets may be free standing or wall mounted as required. For free standing cabinets, front and back access door shall be provided. For wall mounted cabinets, front access doors only shall be provided.

4.3 Construction

Each panel shall be fabricated of ASTM A366 sheet steel, minimum 2.3 mm thick. Each panel shall be a completely enclosed rigid, freestanding structure, suitably braced, stiffened and reinforced to withstand without damage, all stresses incidental to shipping, installation, and operation. The panel structural framework shall be fabricated to channels or angles not less than 6 mm thick. Welded construction shall be used throughout.

Means shall be provided so that the panel can be moved safely to position without straining or distorting the structure. Removable lifting eyes or angles shall be provided.

The length and depth of each panel shall be as required by its service. The height shall not more than 2200 mm.

Rounded or “broken plate” construction shall be used on all front or side vertical corners. All joints between adjacent plate sections shall be butt-welded and reinforced on the back if necessary. Base angle of panels shall be continuous.

The workmanship shall be such as to result in a neat appearance outside and inside with no welds, rivets or bolt heads apparent from the faces and sides of the panels and with all exterior surfaces finished true and smooth. The cut out dimension tolerances shall be such that the edge of the cut-out is not visible with the equipments installed. All cut-out edges shall be square and smooth.

Internal brackets shall be provided for mounting accessory equipment. Structural steel framework shall be provided for supporting incoming cables, wiring, and other appurtenance.

The panel face, top and sides shall be built and reinforced in such a way that any deflection due to the weight of instruments, interconnecting wiring, piping etc. shall not be more than 1.5 mm.

Temporary bracing shall be provided at doorways to keep panels rigid up to the time of setting in place in the field.

The Panels shall be designed to receive cables from bottom or top as needed or as advised by the Owner. In addition, wall mounted panel may receive cables from the sides. Removable gasketed cover plates shall be provided for drilling for conduits and cable entrance fittings.

All cables which enter to the panel shall be adequately supported.

All panels except for wall or skid mounted types shall be furnished complete with separate bases. The bases shall be so designed that they can be set, leveled and grouted in place before the panel is set upon them. If anti-vibration mounts are required, the Contractor shall furnish them.

All Control panels shall be furnished with channel base, for the setting of exact horizontal level.

Wall mounted panels shall be furnished with mounting frames and hardware for wall mounting.

Skids on which panels are mounted shall include suitable panel bases either integral with the skid structure or designed for rigid attachment to the skid. If the equipment mounted on a skid is a source of vibration, panels mounted on the skid shall be provided with anti vibration mounts.

4.4 Painting and Finishing

4.4.1 General

The Contractor's procedure for preparing the panel for painting and the number and method of applications shall stipulate the following:

- a. All metal surfaces shall be cleaned and wire brushed to remove adherent scale and rust prior to applying the priming coats.
- b. The Contractor shall submit a detailed painting procedure for all the panels and other equipment in his scope for the Owner's approval. The

procedure shall include surface preparation, paint process and physical characteristic, application and type of the paint to be used.

- c. All welding on the panel structure shall be completed prior to applying the priming and finish coats.
- d. All surfaces shall be cleaned by vapor degreasing or washed with petroleum solvent, and painted (prime and finish) at the manufacturing plant per manufacturer's standard. However, the paint shall be suitable for the environmental conditions at site.

4.4.2 Finishing for Indoor Panels

Exterior surfaces shall receive two coats of both lacquer proof primer, and a finish coat of manufacturer's standard lacquer.

Exterior surfaces shall be finished to give a high quality, durable, semi gloss surface. The Contractor shall propose the color of exterior surface to the Owner after contract award for Owner approval.

Interior surfaces of the panel shall receive a priming coat and a finish coat of anti condensation white paint.

4.4.3 Finishing for Outdoor Panels

All surfaces shall be properly cleaned and prepared to receive prime and finish coatings of epoxy paint.

The Contractor shall propose the color of exterior surfaces to the Owner after Contract Award for Owner approval.

Interior surfaces shall receive a priming coat and a finish coat of white paint.

4.5 Door, Opening and Covers

Access doors shall have a hinge assembly that shall permit door removal for un-restricted access to the internal equipment.

Access doors of walk-in type panels shall be flush-mounted type with door closures and concealed hinges. Panic door operators shall be furnished on the inside of the doors.

All doors shall be of ASTM A366 (or equipment standard) steel and shall have locks, handles and 3-point latches.

Doors shall be capable of at least a 1300 swing with 900 from the closed position and shall be provided with a hold-open mechanism.

Removable access panels shall be flush type, properly braced and stiffened and provided with flush-mounted latching devices to facilitate easy removal.

Components mounted inside the panel shall maintain a 50 mm minimum clearance from the nearest part of any door or component mounted there-on.

Contractors shall provide necessary openings and tapped holes in the panels for all panel mounted devices, relays, control switches, lights, etc.

Where the future addition of components is specified, the Contractor shall also provide openings, cover plates, tapped holes, and rustproof mounting hardware as required to accommodate the future additions.

Openings with gasketed cover plates shall be provided for cable entry.

4.6 Terminal Blocks

Terminal blocks shall be provided for the termination of all cables and wiring entering the panels except for prefabricated plug-in cables for which suitable sockets shall be provided.

Terminals associated with individual items of equipment shall be grouped and terminated at separate terminals for convenient cable connection.

Terminal blocks shall be mounted vertically on brackets extending out from the rear of the panel. The top or side of any terminal block shall not be closer than 150 mm to the top of panel. Horizontal distance between terminal blocks shall be a minimum 200 mm. The terminal block shall subject to Owner review and acceptance.

A minimum of 15 % spare terminals shall be provided. Only one wire per terminal shall be used on the outgoing side of terminal blocks. The Contractor shall provide any common connection required on the panel side of the terminal block.

4.6.1 Terminals for Power Wiring

Terminal blocks shall be of heavy-duty molded type. Screw type terminals with washer head screw shall be provided for terminating all wiring and inter-connecting cables. Barriers between terminals shall be of such height that terminal screws or wires lugs will not project above the barriers. Space between barriers shall be sufficient to accommodate the wire lugs which shall not exceed 9.5 mm in width. Terminals shall be clearly and permanently identified.

Terminal blocks shall be rated 600V, 30 Amp. The terminal blocks shall be provided with a designation strip for identification of each incoming or outgoing wire. The strip surface shall be suitable for permanent notation and shall be mounted such that it cannot be replaced in any but the original orientation.

4.6.2 Terminals for Instrument Wiring

Terminal blocks shall be molded type with screw clamp type terminals.

For shielded field cables, a separate terminal shall be provided for each shield drain wire.

Wires for external connection shall not be terminated on PCB (printed circuit boards).

4.7 Wiring

4.7.1 Wire

All shop wiring shall be Class 2 or 5, as required, stranded copper wires except for connections between fixed and hinged sections which shall be Class 6 extra flexible wire.

Wires shall have non flame propagating 90°C XLPE (cross linked poly ethylene) insulation.

The wire size and voltage ratings shall be as follows:

- a. Power wiring shall be rated 600 V with minimum wire size of 6.0 mm².
- b. Control wires shall be rated 600 V with minimum wire size of 2.5 mm².

- c. Minimum wire size from secondary winding of current transformer shall be 6.0 mm².
- d. Instrument low-level signal wiring shall be rated 300 V with minimum wire size of 1.0 mm². For special requirements with wire size smaller than 1.0 mm² application prior approval from the Owner is required.

The wire between receiving devices and terminal blocks for direct thermocouple connection shall be of the same type as the field connected thermocouple extension cable.

4.7.2 Wiring Arrangement

Wires and wire groups extending from device to device, or from devices to terminal blocks, shall be installed in wire-ways and/or packets in neatly formed bundles securely clamped or tied together with cable ties attached and secured to the panel framework with wire cleats.

Where wiring is installed in wire-ways, the edges of the cut-outs through which the wires pass shall be suitably protected or grommeted to protect the insulation from cuts or nicks.

Shielded instrument cables carrying low-level signals shall be in bundles or wire-ways separate from those for power or control cables.

Wiring shall not be routed in such manner that it will prevent blank panel space from being used for the addition of future equipment.

All wiring, non-metallic components and accessories such as terminal blocks, wire-ways, wire cleats, cable ties, etc. shall be manufactured from “non burning” material as defined by IEC Standard.

Adhesive backed wire bundle grips are not acceptable for supporting wire extensions.

Bends in wiring shall be carefully made in such a manner that the insulation or cover is not damaged. Care shall be used in removing insulation from the wire, so that the wire will not be cut or nicked.

Supply circuits for AC service to instruments shall be run in wire-ways provided with receptacle for polarized locking plugs, as required. A convenience

outlet with plug and flexible extension cord shall be furnished for each instrument not otherwise provided with built-in means of disconnecting the AC supply circuit.

4.7.3 Wiring Termination

All internal and external wiring terminations shall be made with compression type (solder-less) lugs except for thermocouple extension cables and for sockets for plug-in cables.

The maximum number of conductors connected to any one terminal point on a device or terminal block shall be two.

Shielded cable drain wires and shield tapes shall be fully insulated and terminated at terminal blocks. The shield shall be grounded at one end only.

Grounding of shield wires for un-grounded thermocouples shall be at or near the thermocouple lead terminal boxes. Instrument cable shields shall be grounded at the panel.

All incoming power terminals shall be clearly identified in a manner distinctly different from all other terminations for safety in maintenance.

All internal wiring shall be clearly and permanently marked at both termination ends with circuit designations.

All internal wiring shall be terminated on one side of terminal blocks so that the field side is left clear for field cable and wire connections.

All wiring shall be installed exactly as shown on the wiring diagrams.

All spare contacts from each auxiliary relay and control switch shall be wired to terminal blocks.

The wire insulation shall be removed without nicking or cutting the conductor. The wire shall be firmly inserted into the lug and crimped with a ratchet type or positive action air/hydraulic tool.

All wires or wire terminals projecting beyond the terminal block shall be insulated.

All bare bus and power connections shall be shielded to prevent accidental contact by persons.

4.8 Name Plates

Nameplates shall be installed to designate the service of all switches, instruments, and meters, fuse blocks, relays, amplifiers, etc both internally and externally located on the panels. Standard nameplates included with instruments will not be considered adequate. The Contractor shall provide separate nameplates bearing the service description, instrument number, and calibrated range.

Nameplates shall be made of not less than 1.6 mm thick stainless steel engraved with black letters on white background. Lettering shall be of the commercial Gothic type and in upper case (capital only). Letter shall be 5 mm high with 28 letters or spaces per 100 mm. Nameplates shall be fastened with two rust resistant machine screws. Any instrument, which is mounted on the panel front, shall have nameplates on both the front and rear of the instrument. The instrument nameplates mounted on the front of panel shall be engraved with English language legends. The nameplates shall be located above the instrument for ease of reading. Deviation from these requirements shall be with prior approval by the Owner.

The nameplates for the rear of the instruments and devices shall be engraved with English and Indonesian language legends. Each panel and enclosure shall also be furnished with nameplates on the front and back with English and Indonesian language legends. A detailed nameplate list with service legends shall be submitted for Owner's review. Refer to Section 2, Sub-clause 2.8.3.

4.9 Panel Mounted Devices

4.9.1 General

Front-of-panel mounted devices shall not be located lower than 760 mm or higher than 1700 mm above the floor level (with exception of annunciators) without prior approval by the Owner

Devices inside the panel shall not be located lower than 500 mm or higher than 1700 mm above the floor level without prior approval by the Owner.

Instrument devices exerting strong cantilever forces on panel steel shall be adequately supported to remain plumb and true within 1.5 mm of the desired vertical and horizontal alignment.

Devices shall be mounted with adequate spacing allowed for wiring and/or piping connection so that instrument removal, replacement and servicing may be accomplished without interruption of service to adjacent devices or without unduly disturbing other wiring and/or piping connections. All devices shall be so located that terminals and adjustment devices are readily accessible without use of special tools and with terminal marking clearly visible.

1. Generally local controls and monitoring shall be performed using conventional controls (i.e. pushbuttons and lamps) or on local small VDUs directly attached to the relevant DCS or PLC. The function of annunciators shall be same as that of hard wired annunciators described below.
2. All necessary signals including annunciators shall be transmitted to the Plant Information Management Station through the relevant DCS, RIO, PLC, or Gateway.

If a hard wired control system is provided, annunciators, control switches, push buttons, instruments and alarm/indicating lamps shall be located on the front of the panels. Relays and amplifiers etc. shall be located within the panels. Lamps shall be in the form of LEDs – filament lamps are not allowed.

4.9.2 Annunciators

If a hard wired annunciator system is provided, it shall be a back lighted window type and connected to the DCS for plant information management. A hard wired annunciator system shall have following features:

- a. Translucent 50x75 mm windows with white background and black engraved lettering. The proposed lettering size and arrangement shall be to the approval of the Owner.
- b. Brightness of the window areas shall be sufficient to provide ample contrast. A minimum of two (2) lamps shall be provided for each window. The lamps shall be replaceable while the equipment is in service.
- c. Dull black finished sheet steel enclosures suitable for flush or semiflush mounting. All equipment in the cabinet shall be readily accessible for

maintenance without removal of the cabinet connection. Hinged doors or removable panels shall be provided for this purpose. All internal wiring shall be complete to terminal blocks in accordance with clause 4.6.2. The spacing between rows of terminal blocks and between a row of terminal blocks and other equipment on the cabinet shall be minimum of 75 mm.

- d. Logic circuitry, power supplies and associated components of the solid-state type shall be installed in the annunciator cabinet.
- e. “Silence”, “Acknowledge”, “Test”, and “Reset” push buttons shall be mounted separately from the annunciator on the front of the panel at a height within reach of the Operator.
- f. Inputs shall be suitable for use with alarm contacts which are either close or open in the alarm condition. Provisions shall be included to change an input from one to the other in the field without a wiring change. Spurious electrical noise voltages shall be rejected and there shall be no false signals due to contact bounce up to 30 millisecond either on alarm or return to normal. Alarm contacts shall be monitored at 110 V DC which shall be provided by the annuunciators.
- g. Protection to prevent surges and over-voltage on the input circuits or in the power supply from damage or from interfering with the correct operation of the equipment shall be provided for semiconductor elements. The system shall be designed with a surge withstand capability in excess of the requirements indicated in ANSI C37. 90a.
- h. Audible output device rated for continuous operation shall be at a 10 db above ambient noise and a minimum sound output of 93 dbA. The audible output device shall be powered from the annunciator power supplies and shall be suitable for surface mounting. The audible output shall be timed so that it stops sounding if it is not acknowledged within 10 minutes. The timer shall be adjustable from 5 to 60 minutes.

- i. Annunciator sequence shall be as follows

Condition Trouble	Contact	Window Lamps	Audible
Normal	Normal	Off	Silent
Alarm	Abnormal (Maintained or Momentary Contact)	Flashing	Sounding
“Silence”	Normal or Abnormal	Flashing	Silent
“Acknowledge”	Normal or Abnormal	Steady On	Silent
“Test”	Normal or Abnormal	Flashing	Sounding
“Reset”	Normal	Off	Silent
	Abnormal	Steady On	Silent

- j. Operation of the “silence” push button prior to operation of the “acknowledge” button for an abnormal condition shall not negate the operation of the audible signal for a subsequent abnormal contact operation other than for points already in alarm.
- k. Contacts to signal “System Trouble” to the plant computer shall be provided. The contacts shall open and reclose if a further alarm occurs before the previous one has been cleared.

4.9.3 Relay Protection and Monitoring Equipment

The relay protection and monitoring equipment shall be accordance with part 4, Technical Specification E-001.

4.9.4 Indicating Lamps

The indicating lamps shall be accordance with Part 4, Technical Specification E-001.

4.9.5 Control Switches

The control switches shall be accordance with Part 4, Technical Specification E-001.

4.9.6 Push Buttons, and Push Button Station

The push button, and push button station shall be accordance with Part 4, Technical Specification E-001.

4.9.7 Sound Power Jack System

The sound power jack system shall be in accordance with Part 4, Technical Specification E-001.

4.9.8 Auxiliary instrument Transformer

The auxiliary instrument transformer shall be accordance with Part 4, Technical Specification E-001.

4.10 Interior Lighting

220 VAC fluorescent interior lighting with switches shall be furnished inside the panel. It shall be connected via suitable fuses to a separate incoming 220 VAC power supply from a lighting cabinet. The light fixture shall have a degree of protection against electrical shock of Class I or Class II according to IEC 60598-1 Class I fixtures shall be grounded.

4.11 Convenience Plug

220 VAC, two-pole plug receptacles with side earthing contacts shall be furnished inside each panel spaced at a maximum of 2 meters apart. Minimum number of receptacles shall be two (2). They shall conform to design C2a of IEC 60083. They shall be connected via suitable fuses directly to the incoming 220 VAC power supply.

All panels and cubicles shall be fitted with an approved telephone jack socket which shall be wired down to terminal blocks for connecting to a maintenance telephones system. Telephone jack socket shall be fitted on the front left hand side of all panels or cubicles.

4.12 Grounding

All panels shall be equipped with copper ground bus of not less than 7 x 25 mm cross section extending the entire length of the panel. The ground bus shall be securely anchored to the-panel so as to effectively ground the entire structure and have provision at each end for connecting to the grounding inter-connection between shipping splits.

Each instrument, current transformer, and relay circuit ground shall be a single insulated copper wire not less than 2.5 mm² that shall be run independently to the ground bus and fastened thereto with a suitable terminal plug.

The cases of all relays, recorders, etc are considered to be effectively grounded through the panel structure.

4.13 Ventilation Fans

If forced ventilation is required to dissipate heat generated internally in indoor panels, ventilation fans complete with intake filters and all necessary grilles shall be furnished. An alarm contact shall be provided to indicate ventilation malfunction.

Each fan shall be of sufficient capacity to maintain 0.5 millibar positive pressure within the panel and limit the air temperature rise to 50⁰C. An alarm contact for malfunction of ventilation fans shall be provided.

4.14 Anti Condensation Heaters

All panels shall be equipped with anti condensation heaters in accordance with Part 4, Technical Specification E-001.

4.15 Piping and Pneumatic Connection

Contractor shall furnish and install all piping and/or tubing connections within the panels as required for all panel mounted instruments, control stations, relays, solenoid valves, braces, supports etc. as required. Full information about piping, tubing and fitting sizes used shall be provided to the Owner.

All pneumatic supply, control and signal lines shall be soft annealed copper tubing, valves shall be brass or bronze and fittings shall be brass or bronze flared or compression type. Tubing bend shall be formed with mechanical benders. Flattened, kinked or wrinkled bends are not acceptable.

Tubing and piping shall be installed in such a manner as to avoid interference with panel mounted equipment removal or access to the equipment for maintenance purposes.

Shutoff valves mounted in pneumatic lines shall be adequately supported so as to prevent undue torque from being applied to the tubing with the opening or

closing of the valves. The valves shall not be solely supported by the tubing. Valves shall be easily accessible from rear of panel. Shutoff valves for pneumatic loading lines shall be mounted at the piping terminal plate or bar.

Auxiliary components such as pressure switches, solenoid valves etc. shall be rigidly supported from the control panel assembly framing or wall plate.

Piping terminals shall terminate on the panel top plate pointing “up”.

Each piping terminal connection, on both the inside and outside of the panel, shall be provided with a metal tag or suitable permanent marking strip as reviewed by the Owner to indicate the instrument tag numbered being served by the connection, as well as the instrument port designation.

Tubing connections on the piping terminal plates shall be arranged in neat and orderly manner with space provisions to allow for access to make and break termination connections.

4.16 Hazardous Location

Local control panels located in hazardous location shall be in accordance with NFPA 70 Article 500 and IEC 60079.

4.17 Surge with Stand Capability

The equipment shall be tested in accordance with requirement of ANSI C37.90. Following this test the equipment shall be given a calibration test to determine that it has not been damaged by the surge withstand capability test.

The Contractor shall make tests on completely assembled system and shall provide Owner data on above tests.

5. TESTING

All panels shall be completely assembled and tested in the factory.

All tests and guarantees shall be made in accordance with applicable IEC or ANSI Standard and in Technical Specification E-001.

a. The Electrical Tests

The electrical tests shall demonstrate adequacy and integrity of the isolation, continuity and accuracy of all internal wiring. The test shall include point-to-point continuity test and electrical insulation test (Note: Megger or other voltage tests shall not be applied to any coaxial shielded cables or solid state components). The Contractor shall be responsible for proper protection of instruments and devices that may be damaged by high voltage tests.

Tests and checks shall be made by Contractor before shipment to insure that all electrical equipment and wiring furnished and installed by Contractor is in proper operating condition and that the wiring is in exact accordance with the wiring diagram and that all wiring will function as intended.

The wiring shall capable of withstanding one minute test at 1500 VAC.

b. Hydrostatic Test

Demineralized water shall be used for all hydrostatic testing and the test shall be in accordance with the requirements of ANSI B31.1.

All tubing and piping connections, welds, and valves, stems and seats shall be checked for leaks. Detectable leakage shall not be permitted. Leaks shall be repaired and the equipment retested.

During the hydrostatic test, all instrument devices shall be disconnected beyond the shutoff valves. Valves shall be placed in an open condition and the line or valve capped. This is intended to test the valve bodies only and prevent damage to the valve seals.

c. Pneumatic Test

All pneumatic piping, tubing, valves and fittings shall be tested with 10.5 bar nitrogen in accordance with the requirements of ANSI B31.1, with the final connection at the instruments disconnected and plugged. Each joint shall be inspected for leaks using a soap bubble-type test with chlorine or chloride free fluid. Detectable leakage shall not be permitted. Leaks shall be repaired and retested.

6. QUALITY ASSURANCE /QUALITY CONTROL

- a. Quality Assurance/Quality Control shall conform to the requirements of Part 4, Sub-section 2.7.
- b. The Quality Control procedures as specified in Part 4, sub-section 2.7 shall include but not limited to the following:
 - i. Continuity and Wiring Tests
 - ii. Insulation and High Potential Testing
 - iii. Packaging and Shipping
 - iv. Welding
 - v. Cleaning and Painting
- c. The Quality Assurance/Quality Control documentation as specified in Part 4 Sub-section 2.7 shall include, but not limited to the following:
 - i Shop Test Reports
 - ii All other documentation required by applicable codes and standard.

(Last page of I - 002)